

Presenting Research Plans and Proposals

LS100 — Module 00A · Research Plans & Proposals

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This guide is about the first presentation for the LS100 students — not the written proposal, which is covered separately. It's for the 15-minute talk you give as a group at the very start of the semester.

Because you haven't run the study yet, this talk isn't about results. It's about showing your plan holds together, and getting feedback on it while you have time to act on it. Build the talk to invite questions, not avoid them. Keep it to a maximum of **four slides** — here's what may go on each one, and how to present it well.

1. YOUR FOUR SLIDES

Slide	Content
1	The overarching question — and why it matters
2	Your specific aims and hypotheses
3	Methodology and data — how each dataset answers an aim
4	Possible outcomes, and back to the bigger question

This mirrors how NIH grant applications open: a one-page “Specific Aims” section that states the problem's significance, lists the aims and hypotheses, and closes on expected impact ((National Institute of Allergy and Infectious Diseases (NIAID), n.d.)).

2. 1. DEFINING AND FRAMING THE PROBLEM

Slide 1

- Define the problem scientifically: state it as one specific, testable question, not a broad topic. (“What drives X” is a topic; “Does A predict B in Y” is a question.)
- Say why this is interesting to you, and why it might matter to a broader audience — especially if the full research would need substantial funding.
- Keep it to one sentence your audience could repeat back to you. If they can't, it isn't ready for slide 1.

- Ideally, you would frame the research question with naming the gap — what’s known, and what isn’t — that your question sits in. But for this presentation, this depends on how much time you want to allocate to this slide.

For the talk: You might open this as a short story, not a list of facts or questions — what’s known or agreed on (the “and”), the tension or gap (the “but”), and what you propose to do about it (the “therefore”). One well-built sentence like this can carry the whole slide ((Olson, 2015)).

= 2. Breaking down the problem: specific aims and hypotheses

Slide 2

- To solve the problem, what related specific questions are needed to be answered, or steps needed to be taken, either in sequence or in parallel? This could be answering one particular question, testing one particular hypothesis, or developing a tool, or a statistical or machine learning model.
- State each as a specific aim: what you’ll do, and what you expect to find. Each question or step should have a clear hypothesis or outcome. Think of hypothesis as a specific, falsifiable answer — “if [X], then [Y], because [mechanism/ logic]” is a good template.
- Check whether each aim is doable in the time and resources you actually have. If a later aim depends entirely on an earlier one succeeding, say so and name it as a risk.
- Write down the questions, hypotheses, steps, and possible outcomes in a logical sequence. This process is iterative, so you may need to revisit it and add, remove, or reorder items — that’s expected.

For the talk: Keep this to two aims. Even NIH grant writers are told to limit their aims to avoid looking unfocused rather than ambitious ((National Institute of Allergy and Infectious Diseases (NIAID), n.d.)) — the same holds on one slide.

= 3. Focusing on one achievable goal: Methodology and data structure

Slide 3

- Next, focus on the first question or step. Is it doable within your available time and resources?
- Name every dataset you’ll need and its type, and draw a direct line from each dataset to the aim it answers.
- Say where the data comes from, what equipment or access it needs, and how long it will take to collect. As a rule of thumb, **consider budgeting about a third of your available time for collecting data.**
- Name the proxy you’re using for anything you can’t measure directly (e.g., a click count standing in for “engagement”), and say in one sentence why it’s a

reasonable stand-in — and one way it could be wrong (a threat to what methodologists call construct validity: (Cronbach & Meehl, 1955)).

- Note, briefly, any transformation the raw data needs before analysis, and the statistical or computational method you'll use.

For the talk: You may show this as a diagram or infographics, for instance, a graphics showing “from a specific aim, arrows to each dataset and method”. A figure someone else could present without you standing there is doing its job; if it takes more than a few seconds to read, simplify it ((Midway, 2020)). Watch for the usual traps: a bar chart with a non-zero y-axis, two unrelated variables on dual y-axes, or a pie chart — all common ways figures mislead rather than clarify ((Nguyen et al., 2021)).

= 4. Possible outcomes and the bigger picture

Slide 4

- State the outcome that would tell you the analysis worked — a hypothesis rejected, or a pipeline or model produced.
- Connect this specific aim back to the larger theoretical, practical, or societal question you opened with, and name the next question that follows from it.
- Say what you'll do with the feedback from this talk. This slide is a draft, not a conclusion.

For the talk: Close by answering the “who cares” question you opened with — bookend the talk. Compressing a research question and its stakes for a general audience in a few minutes is a real, transferable skill.

= Delivering it well

- Rehearse out loud against a timer. Four slides in 10 minutes is roughly 2–2.5 minutes each, including time to talk through a figure.
- When someone asks a question, repeat or rephrase it before you answer — it confirms you understood it, and makes sure the room heard it too.
- Build the talk to invite questions, not close them off: ask yourself what you want your audience to ask, not just what you need to tell them ((Heath & Heath, 2007)). The more engaged your listeners are while you talk, the more they understand and remember afterward ((Stephens et al., 2010)).

3. BEFORE YOU PRESENT: A QUICK SELF-CHECK

- Slide 1: Can someone outside the behavioral sciences follow your one-sentence question?
- Slide 2: Are your aims specific and falsifiable, not just topics?
- Slide 3: Does every dataset trace to an aim, and does the timeline add up?

- Slide 4: Have you named the outcome that tells you it worked, and closed the loop on “why it matters”?
- Have you timed a full run-through, out loud, at four slides?

REFERENCES

- Cronbach, L. J., & Meehl, P. E. (1955). Construct validity in psychological tests. *Psychological Bulletin*, 52(4), 281–302. <https://doi.org/10.1037/h0040957>
- Heath, C., & Heath, D. (2007). *Made to Stick: Why Some Ideas Survive and Others Die*. Random House.
- Midway, S. R. (2020). Principles of effective data visualization. *Patterns*, 1(9), 100141. <https://doi.org/10.1016/j.patter.2020.100141>
- National Institute of Allergy and Infectious Diseases (NIAID). (n.d.). *Write Your Research Plan*. Retrieved <https://www.niaid.nih.gov/grants-contracts/write-research-plan>
- Nguyen, H., Jung, W., & Gupta, N. (2021). Examining data visualization pitfalls in scientific publications. *Visual Computing for Industry, Biomedicine, And Art*, 4(1), 27. <https://doi.org/10.1186/s42492-021-00073-z>
- Olson, R. (2015). *Houston, We Have a Narrative: Why Science Needs Story*. University of Chicago Press.
- Stephens, G. J., Silbert, L. J., & Hasson, U. (2010). Speaker–listener neural coupling underlies successful communication. *Proceedings of the National Academy of Sciences*, 107(32), 14425–14430. <https://doi.org/10.1073/pnas.1008662107>